



Containerized Syslog Appliance

# Splunk4Rookies - Splunk Connect for Syslog (SC4S)

## Lab Guide

### Overview

This lab guide contains the hands-on exercises for the Splunk4Rookies - Splunk Connect for Syslog (SC4S) workshop and is a companion to the full workshop slides to assist you with configuration file contents and flow.

Before proceeding with these exercises, please ensure that you have a copy of the workshop slide deck, which will help to put into context the tasks you are carrying out.

Download the workshop slide deck: <https://splk.it/S4R-SC4S-Attendee>

### Prerequisites

In order to complete these exercises, you will need your own Splunk instance. The presenter delivering your workshop will give you a unique URL for your event, which will allow you to register and create your own Splunk instance.

Please follow the steps provided by the presenter to get access to Splunk!

### Troubleshooting Connectivity

If you experience connectivity issues with accessing either your workshop environment or the event page, please try the following troubleshooting steps. If you still experience issues please reach out to the team running your workshop.

- **Use Google Chrome** (if you're not already)
- If the event page (i.e. <https://show.splunk.com/event/<eventID>>) didn't load when you clicked on the link, try **refreshing the page**
- **Disconnect from VPN** (if you're using one)
- **Clear your browser cache and restart your browser** (if using Google Chrome, go to: Settings > Privacy and security > Clear browsing data)
- **Try using private browsing mode** (e.g. Incognito in Google Chrome) to rule out any cache issues
- **Try using another computer** such as your personal computer - all you need is a web browser! Cloud platforms like AWS can often be blocked on corporate laptops.

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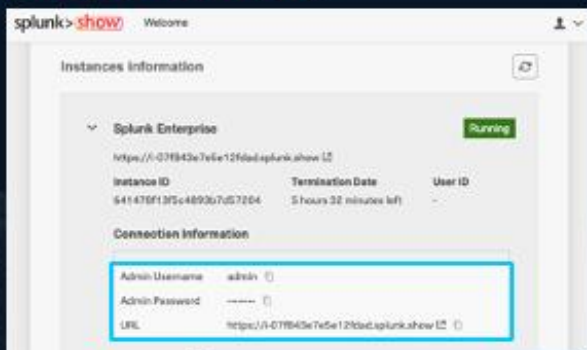
|                                                                                                                |    |
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## Labs: SC4S 101

### Lab 1: Setup Your Environment

# Login to Splunk

Locate your instance URL and credentials  
in the Splunk Show event  
<https://show.splunk.com>

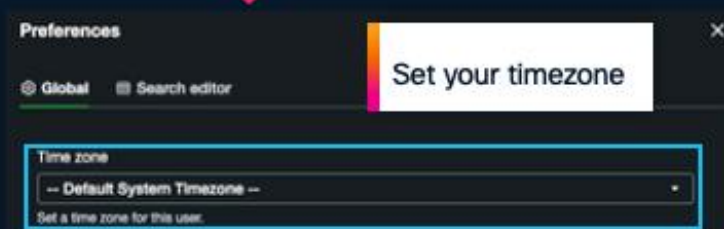
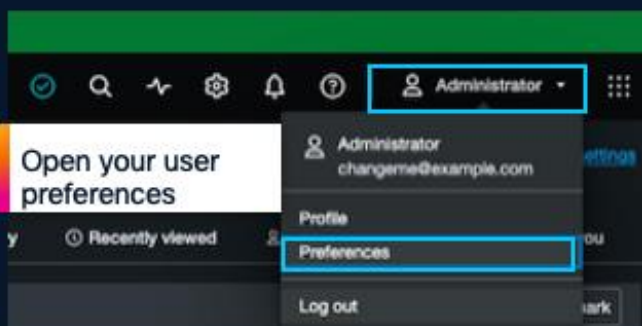
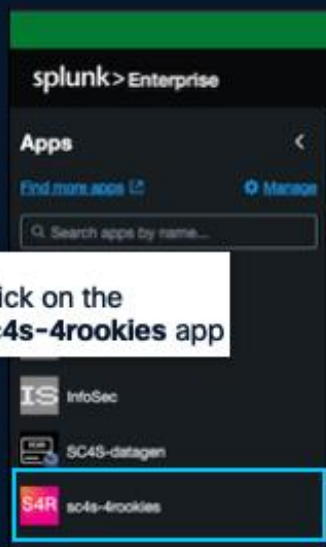


Scroll down the event page  
and expand the **Splunk  
Enterprise** section to view  
your login details

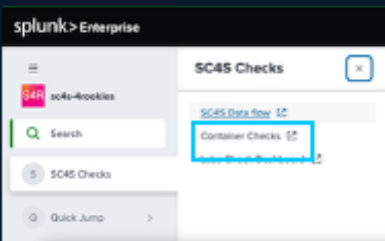
Log in to your Splunk instance



# Login and Set Your Timezone



# Check the Data Flow



Click on **SC4S Checks** and then on **02 - SC4S Container Message Checks**

All inputs are turned off and the sc4s service is stopped after the initial image creation

You may see some historical data or blank tables - this is normal

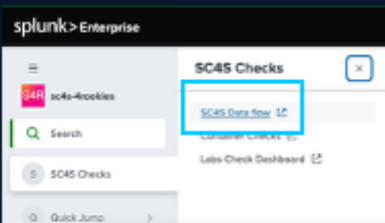
### 02 - SC4S Container Message Checks

Messages from "sc4scontainer" or "sc4sprobe"

Select Time Range: Last 15 minutes

| # | _time                     | sourceType | _raw                                                                                                                            |
|---|---------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 | 2021-09-09T14:17:41+02:00 | sc4sevents | 2021-09-09T02:17:41:728-00:00 sc4s4rookies syslog-ng 164 - [meta.sequenceid="2"] syslog-ng starting up, version="3.32.7"        |
| 2 | 2021-09-09T14:17:36+02:00 | sc4sprobe  | HEC TEST EVENT                                                                                                                  |
| 3 | 2021-09-09T14:17:36+02:00 | sc4sprobe  | HEC TEST EVENT                                                                                                                  |
| 4 | 2021-09-09T14:17:30+02:00 | sc4sevents | 2021-09-09T02:17:30:695-00:00 sc4s4rookies syslog-ng 164 - [meta.sequenceid="276043"] syslog-ng shutting down, version="3.32.7" |

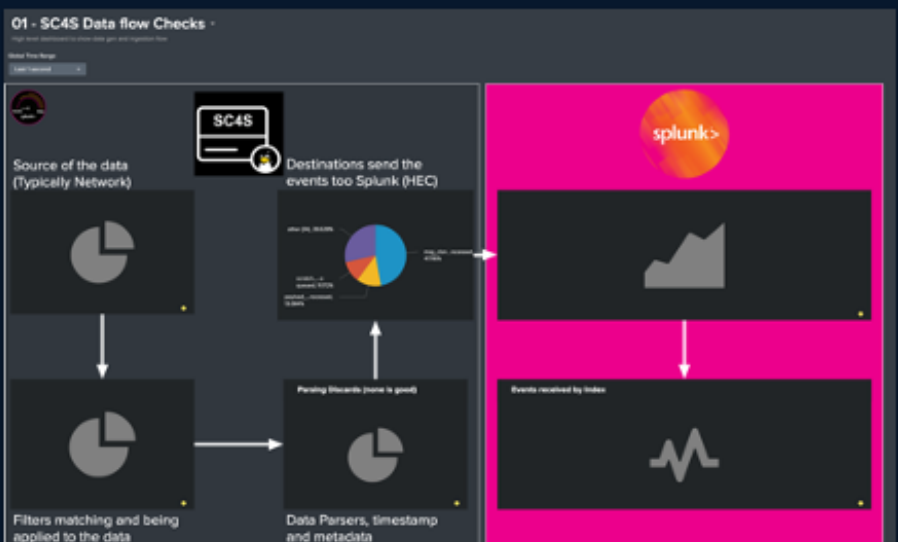
# Check the Data Flow



Click on **SC4S Checks** and then on **01 - SC4S Data flow**

All inputs are turned off and the sc4s service is stopped after the initial image creation

You may see some historical data or blank graphs - this is normal





# Restart the SC4S systemd Service

Restart the systemd service for podman and SC4S by enabling the input, waiting a few seconds, then disabling it again:



Inputs

Manage your data inputs

2 Inputs 10 Per Page All filter

| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Disabled |         |
| sc4s_systemd_restarter | 86400    | default | Disabled |         |

1. Set the toggle to **Enabled**

2. Wait for **10 seconds...**

1. Set the toggle to **Disabled**

**Note:** In environments with CLI access you can also restart the service using the following command:

```
sudo systemctl restart sc4s
```

## Enable the Cisco Datagen

Inputs

Manage your data inputs

2 Inputs 10 Per Page All filter

| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Disabled |         |
| sc4s_systemd_restarter | 86400    | default | Disabled |         |

1. Set the toggle to **Enabled**

Inputs

Manage your data inputs

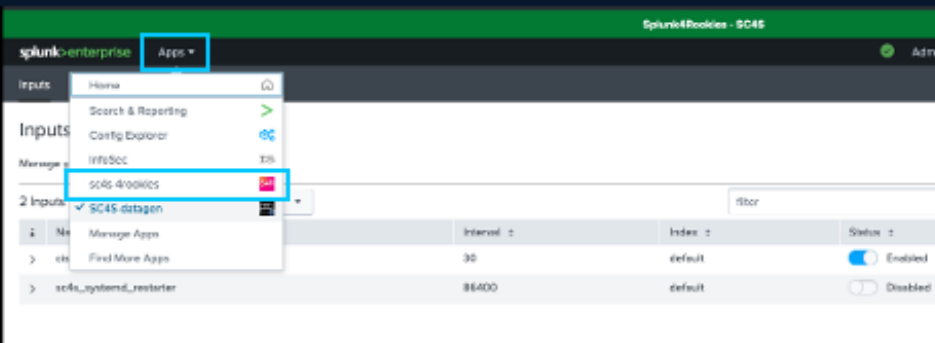
2 Inputs 10 Per Page All filter

| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Enabled  |         |
| sc4s_systemd_restarter | 86400    | default | Disabled |         |

This input will start the cisco datagen - leave this one enabled

Now let's check the data flow dashboard again

# Go Back to the sc4s-4rookies App



The screenshot shows the Splunk Enterprise web interface. The 'Apps' menu is open, highlighting 'sc4s-4rookies'. The main panel shows a list of installed apps:

| Name                   | Interval | Index   | Status   |
|------------------------|----------|---------|----------|
| sc4s-4rookies          | 30       | default | Enabled  |
| sc4s-4rookies-database | 86400    | default | Disabled |

Below the screenshot, a text box says: "Open the sc4s-4rookies app".



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**Check the Data Flow**

Click on **SC4S Checks** and then on **01 - SC4S Data flow Checks**

Now we have started the Cisco datagen you should see a steady flow of data start to populate your dashboard

The arrows indicate the flow of the events through your log paths and out to Splunk over HEC to the indexes

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## Check the Data Flow



Click on **SC4S Checks** and then on **SC4S Container Checks**

Now we have restarted the `sc4s` service you should see some startup messages from `syslog-ng` from inside the container

(If not you may have disabled the sc4s restart input too quickly a few slides back – try that again!)

## Lab 2: Introduction to Search

Q. How would you find DNS events that are NOT .google.com?

Possible answers:

dns NOT \*.google.com

dns query!=\*.google.com

---

### NOT vs !=

For more information on the difference between NOT and != please see

<https://docs.splunk.com/Documentation/Splunk/latest/Search/NOTexpressions>.

---





# Introducing the InfoSec App for Splunk

Download: <https://splunkbase.splunk.com/app/4240>

InfoSec App for Splunk is your starter security pack.

The InfoSec app is free (as in beer) to install and use and is designed to address the most common security use cases, including continuous monitoring and security investigations, as well as a number of advanced threat detection use cases.

All of the components of the InfoSec app can be easily expanded using free security resources available for Splunk like [Splunk Security Essentials](#)

IS InfoSec App for Splunk

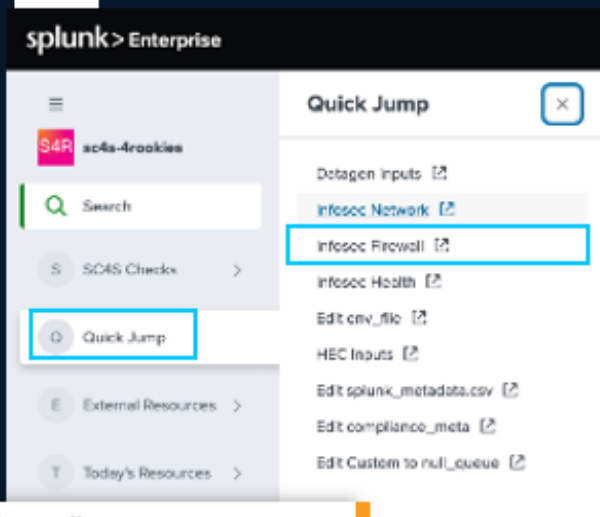
★★★★★ 10 ratings

Splunk Cloud Splunk Built



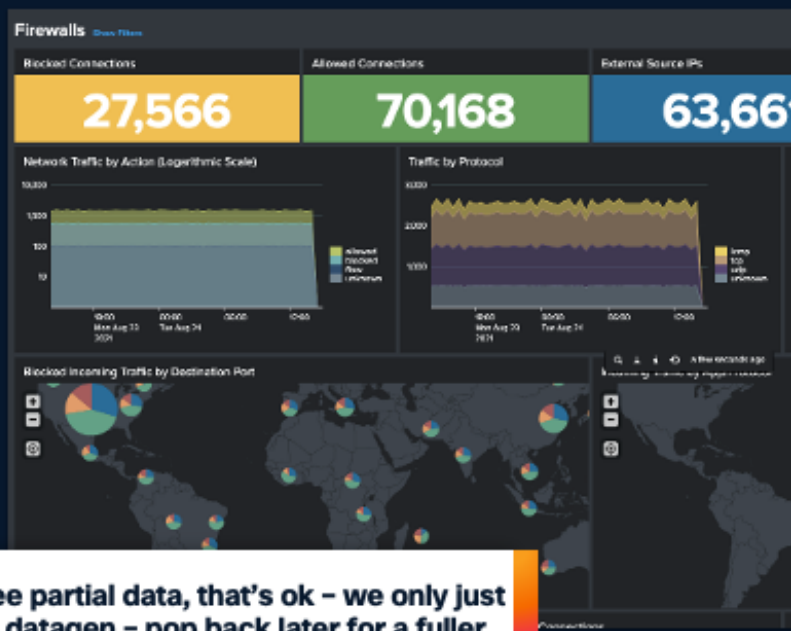


# InfoSec App for Splunk: Examples



## Firewalls

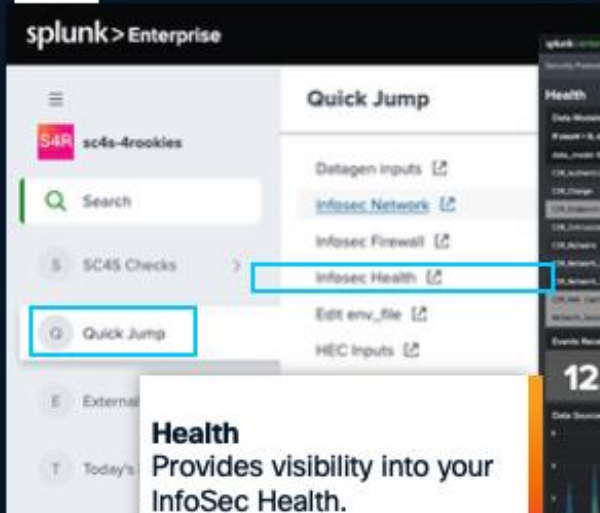
Provides visibility into your firewalls, regardless of vendor or the presence of data from multiple vendors in the same environment.



You may see partial data, that's ok - we only just started the datagen - pop back later for a fuller view



# InfoSec App for Splunk: Examples



## Health

Provides visibility into your InfoSec Health.

InfoSec is built on top of Splunk's amazing CIM data models.



## Lab 4: Config Explorer

# Config Explorer 101

The screenshot shows the Splunk Config Explorer interface. On the left, a sidebar contains a search bar, a 'Quick Jump' button, and a list of configuration categories. The 'Quick Jump' button is highlighted with a blue box. In the center, a 'Quick Jump' panel lists various configuration items, with 'Edit env\_file' highlighted. On the right, a file explorer view shows the path `/sc4s/local/config/app_parsers/` with a 'Filter' button. Annotations with blue arrows point to the 'Recent Files' button, the 'Refresh' button, the 'Filter' button, and the file path. A text box at the bottom left states: 'Config Explorer The basic functions are highlighted here, but it can also do much more than this!'. The bottom of the slide features a copyright notice: '© 2020 Cisco and/or its affiliates. All rights reserved.' and the Cisco logo.

**Config Explorer**  
The basic functions are highlighted here, but it can also do much more than this!

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# Config Explorer 102

It's like Visual Studio Code, but in Splunk!

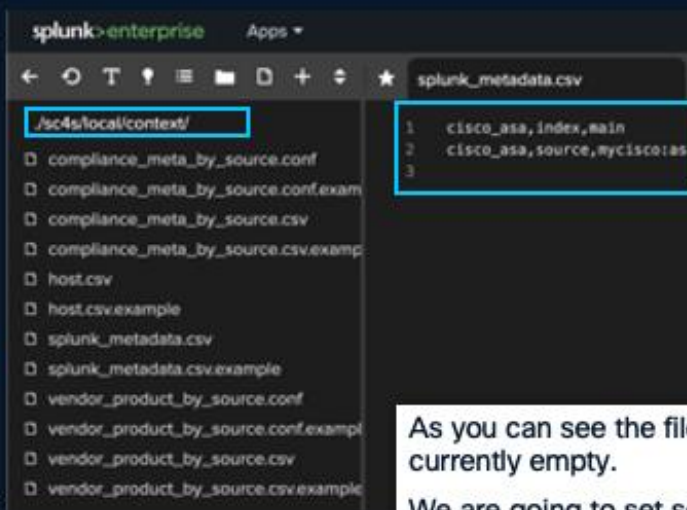
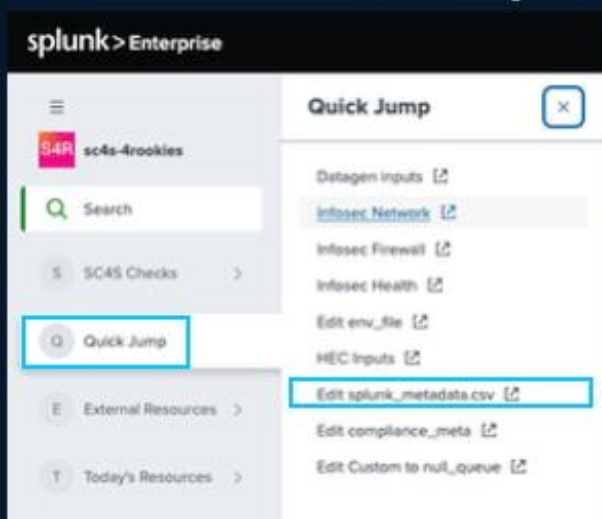
Click on the `env_file` to see the contents:

The screenshot shows the Splunk Config Explorer interface with the `env_file` selected. The left sidebar shows a file tree with `env_file` highlighted. The main area displays the contents of the file, which is a configuration script. The script includes comments and variables for Splunk configuration, such as `SC4S_DEST_SPLUNK_HEC_DEFAULT_URL` and `SC4S_DEST_SPLUNK_HEC_DEFAULT_TOKEN`. The bottom of the slide features a link to learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/configuration/#global-configuration-variables>. The bottom of the slide features a copyright notice: '© 2020 Cisco and/or its affiliates. All rights reserved.' and the Cisco logo.

Learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/configuration/#global-configuration-variables>

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## Edit splunk\_metadata.csv



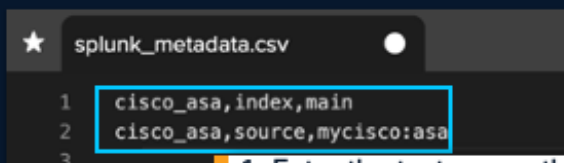
As you can see the file is currently empty.

We are going to set some custom items, restart SC4S and enjoy the show!

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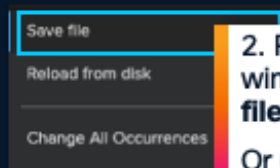
Learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/configuration/#override-the-log-path-of-indexes-or-metadata>

## Edit splunk\_metadata.csv



1. Enter the text as per the screenshot to the left

change:  
index to main  
source to mycisco:asa



2. Right click in the window and select **Save file**

Or use **cmd+S** (Mac) or **ctrl+S** (Windows)

This configuration will take precedence over the default for the **cisco\_asa** filter.

The default values of:

key = cisco\_asa  
metadata = index  
value = **netfw**

key = cisco\_asa  
metadata = source  
value = **cisco:asa**

will become:

key = cisco\_asa  
metadata = index  
value = **main**

key = cisco\_asa  
metadata = source  
value = **mycisco:asa**

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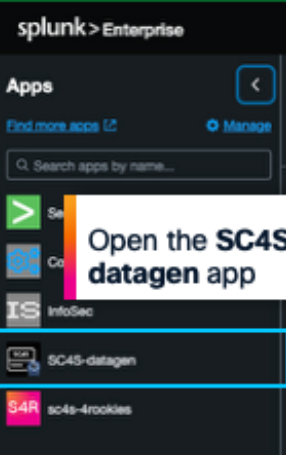
Learn more: <https://splunk.github.io/splunk-connect-for-syslog/main/sources/Cisco/#product-asa-and-ftd-firepower>

Splunk\_metadata.csv contents:

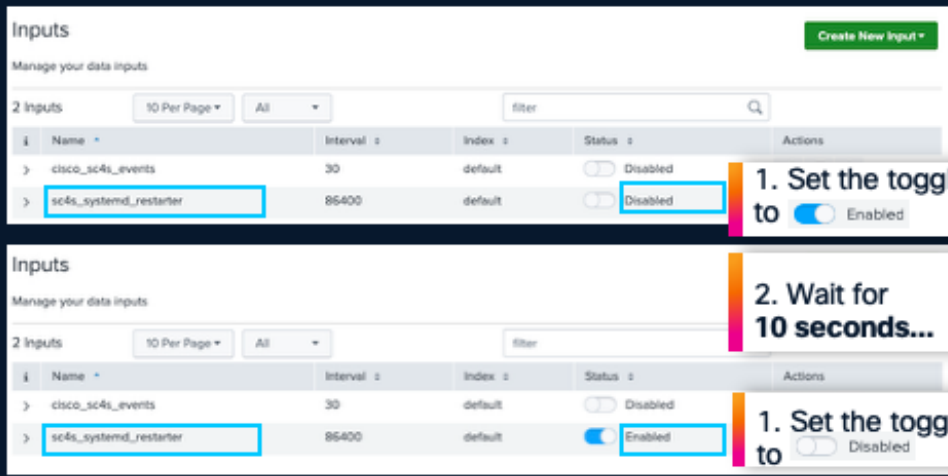
```
cisco_asa,index,main
cisco_asa,source,mycisco:asa
```

## Restart the SC4S systemd Service

Restart the systemd service for podman and SC4S by enabling the input, waiting a few seconds, then disabling it again:



Open the **SC4S-datagen** app



| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Disabled |         |
| sc4s_systemd_restarter | 86400    | default | Disabled |         |

1. Set the toggle to **Enabled**

2. Wait for **10 seconds...**

1. Set the toggle to **Disabled**

**Note:** In environments with CLI access you can also restart the service using the following command:

```
sudo systemctl restart sc4s
```

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Example search to check the changes have been applied:

```
index=main sourcetype=cisco:asa
```



# Check the Changes Have Been Applied

Run the following Splunk search:

```
index=main sourcetype=cisco:asa
```

| i | Time                    | Event                                                                                                                                                                                                                                                                                    |
|---|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 28/08/2021 10:02:19.468 | %ASA-2-106016: Deny IP spoof from (0.0.0.0) to 10.115.120.161 on interface Mobile_Traffic<br>action = blocked   dest = 10.115.120.161   host = 192.168.12.1   index = main   sc4s_container = sc4s4rookies   sc4s_loghost = sc4s4rookies   source = mycisco:asa   sourcetype = cisco:asa |

Our custom index

Our custom source

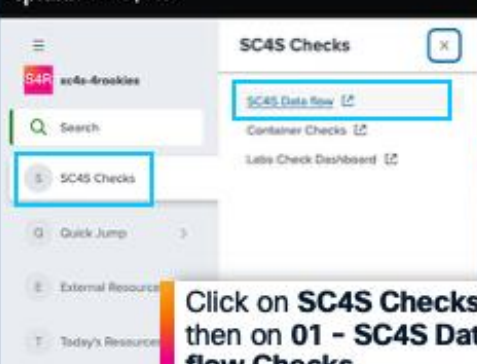
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Learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/configuration/#configure-sc4s-metadata>

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## Check the Index Change

splunk> Enterprise



Click on **SC4S Checks** and then on **01 - SC4S Data flow Checks**



You will notice on the **Events received by Index** graph (bottom right) that we have changed from **netfw** to the **main** index

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Example search to see index changeover:



index=\* | timechart count by index

## Check the Index Change

Run the following Splunk search:

```
index=* | timechart count by index
```



Note the index change here  
when we restarted SC4S

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Learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/configuration/#configure-sc4s-metadata>

Cisco

## Lab 6: Set a Custom Filter

# Filter and Enrich

### compliance\_meta\_by\_source.conf

```

1 filter f_sc4s {
2   host("192.168.12.*" type(glob)) or
3   netmask(192.168.12.0/24)
4 };
5

```

Create **filters** in json format to match the events you want to manipulate

In this case we are choosing our Cisco device

### compliance\_meta\_by\_source.csv

```

1 f_sc4s,fields.compliance,"pci"
2 f_sc4s,fields.datacenter,"DC1"
3 f_sc4s,fields.city,"Auckland"
4 f_sc4s,fields.country,"New Zealand"
5 f_sc4s,fields.lat,"-36.8509"
6 f_sc4s,fields.lon,"174.7645"
7

```

Create **csv** data - this will be added to each event as it flows.

This is a fun example but there are production consequences such as **more disk space used** as fields are **indexed**. Use a **lookup** where possible as it's more efficient!

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Learn more: <https://splunk.github.io/splunk-connect-for-syslog/main/configuration/#override-index-or-metadata-based-on-host-ip-or-subnet-compliance-overrides>

### Compliance\_meta\_by\_source.conf contents:

```

filter f_sc4s {
  host("192.168.12.*" type(glob)) or
  netmask(192.168.12.0/24)
};

```

### Compliance\_meta\_by\_source.csv contents:

```

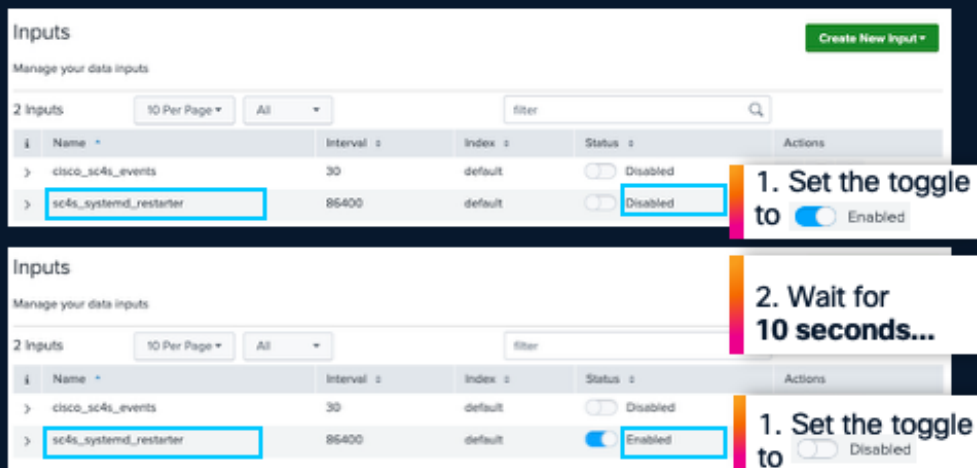
f_sc4s,fields.compliance,"pci"
f_sc4s,fields.datacenter,"DC1"
f_sc4s,fields.city,"Auckland"
f_sc4s,fields.country,"New Zealand"
f_sc4s,fields.lat,"-36.8509"
f_sc4s,fields.lon,"174.7645"

```



# Restart the SC4S systemd Service

Restart the systemd service for podman and SC4S by enabling the input, waiting a few seconds, then disabling it again:



**Note:** In environments with CLI access you can also restart the service using the following command:

```
sudo systemctl restart sc4s
```

## Example search to check events:

index=main

# Check the Changes Have Been Applied

Run the following Splunk search:

```
index=*
```

The raw event is unchanged as we have added fields in flight

Event

%ASA-2-106016: Deny IP spoof from (0.0.0.0) to 10.3.83.54 on interface Mobile\_Traffic

action = blocked

city = Auckland

compliance = pci

country = New Zealand

datacenter = DC1

Our custom fields

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Learn more: <https://splunk.github.io/splunk-connect-for-aviron/main/configuration/#override-index-or-metadata-based-on-host-in-or-subnet-compliance-overrides>

CISCO

Cisco Confidential

# Check the Changes Have Been Applied

```
index=main
```

Our custom fields

SELECTED FIELDS

a action 8

a city 1

a compliance 1

a country 1

a datacenter 1

a dest 100+

a host 1

a index 1

# lat 2

# lon 1

These ones look fun... what can we do with these?

city

1 Value, 100% of events

Selected

Yes

No

Reports

Top values

Top values by time

Rare values

Events with this field

| Values   | Count | %    |
|----------|-------|------|
| Auckland | 2,664 | 100% |

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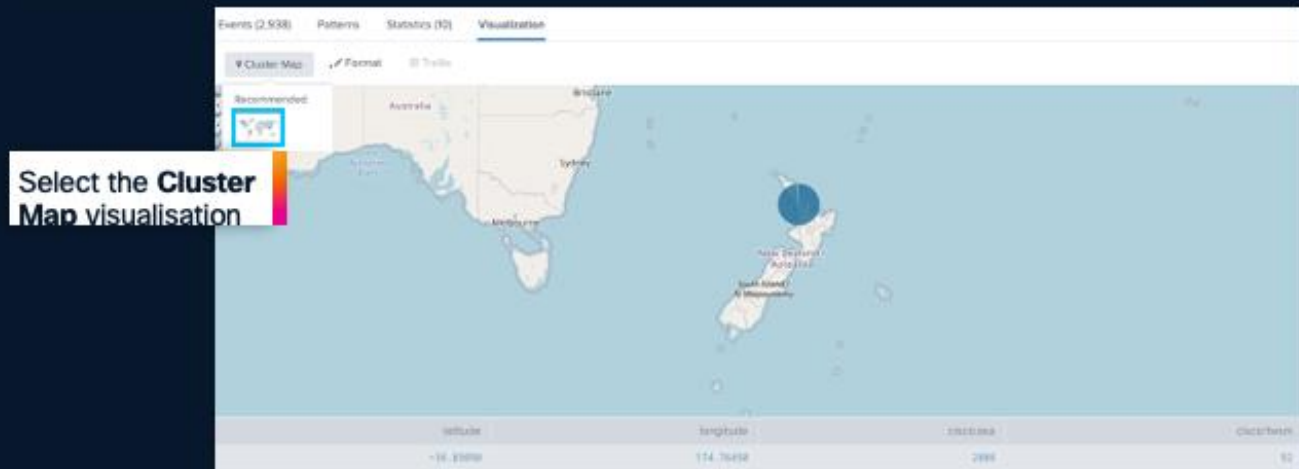
Cisco Confidential

**Example search to check events:**

```
index=main sourcetype=*cisco*
| geostats count by sourcetype latfield=lat longfield=lon
```

## Map It!

```
index=main sourcetype=*cisco*
| geostats count by sourcetype latfield=lat longfield=lon
```



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Learn more: <https://eolink.github.io/eolink-connector-for-sylog/main/configuration/override-index-or-metadata-based-on-host-in-or-subhost-normalization-overrides>

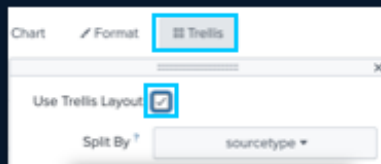
## Lab 7: null\_queue

index=\* host=192\* sourcetype=\*cisco\*

| timechart count by sourcetype

### null\_queue

```
index=* host=192* sourcetype=*cisco*  
| timechart count by sourcetype
```



Enable the trellis layout and split by sourcetype

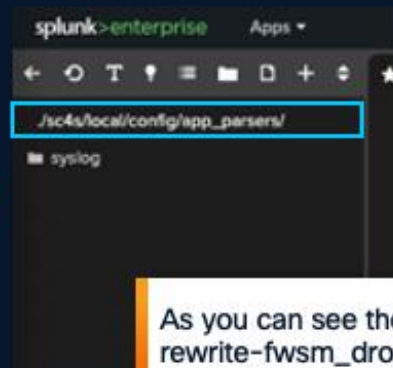
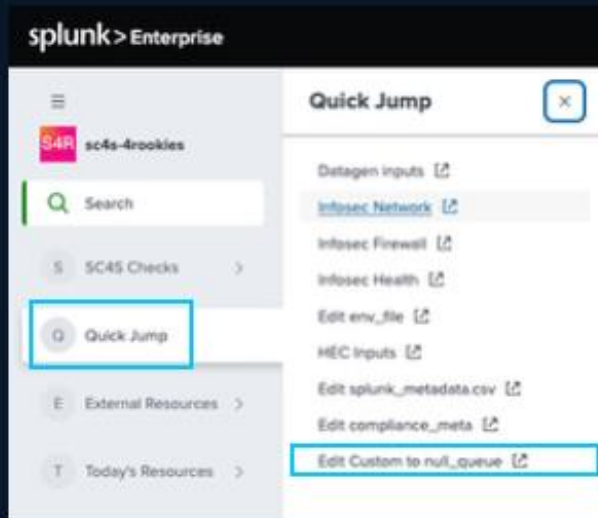


We are currently getting data from:

- asa
- fwsm



# null\_queue



As you can see the dest-rewrite-fwsm\_drop.conf file does not currently exist

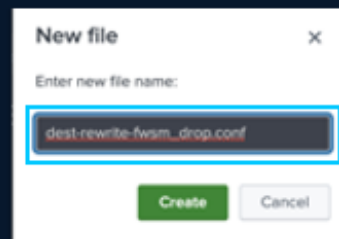
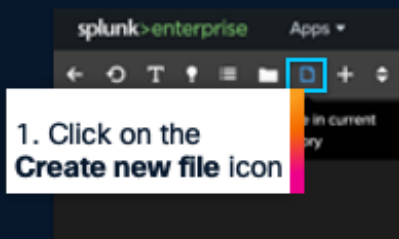
We are going to create the file and its contents

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Learn more: <https://splunk.github.io/splunk-connect-for-syslog/main/configuration/#/connecting-all-data-by-ip-or-subnet>

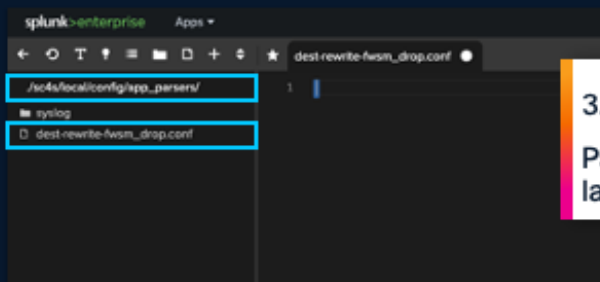
CISCO

# null\_queue



2. Enter the file name:  
**dest-rewrite-fwsm\_drop.conf**  
then click on **Create**.

This file name is arbitrary but make sure it makes sense to you!



3. Double check the file path

Paste the file contents from the lab guide

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Learn more: <https://splunk.github.io/splunk-connect-for-syslog/latest/sources/#filtering-events-from-output>

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## dest-rewrite-fwsm\_drop.conf:

```
# /opt/splunk/sc4s/local/config/app_parsers/dest-rewrite-fwsm_drop.conf

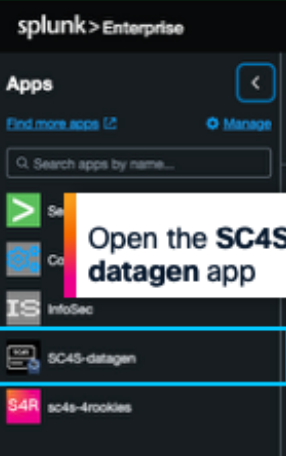
block parser fwsm_drop-postfilter() {
  channel {
    # In this case the outcome is drop the event
    rewrite(r_set_dest_splunk_null_queue);
  };
};

application fwsm_drop-postfilter[sc4s-postfilter] {
  filter {
    "${fields.sc4s_vendor}" eq "cisco" and
    "${fields.sc4s_product}" eq "fwsm"
    # Note regex reads as start from first position
    # Any atleast 1 char that is not a '-'
    # constant '-7-'
    # and message('%[^-]+-7-');
  };
  parser { fwsm_drop-postfilter(); };
};
```

 SC4S-datagen

## Restart the SC4S systemd Service

Restart the systemd service for podman and SC4S by enabling the input, waiting a few seconds, then disabling it again:

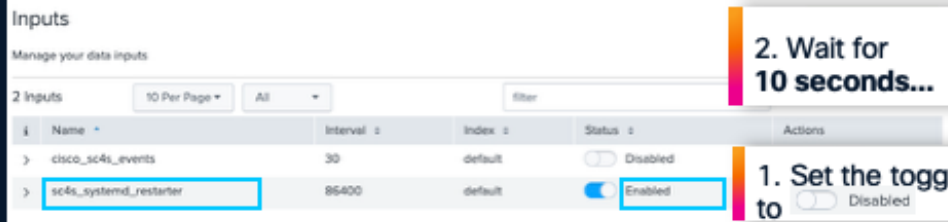


Open the **SC4S-datagen** app



| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Disabled |         |
| sc4s_systemd_restarter | 86400    | default | Disabled |         |

1. Set the toggle to **Enabled**



| Name                   | Interval | Index   | Status   | Actions |
|------------------------|----------|---------|----------|---------|
| cisco_sc4s_events      | 30       | default | Disabled |         |
| sc4s_systemd_restarter | 86400    | default | Enabled  |         |

2. Wait for **10 seconds...**

1. Set the toggle to **Disabled**

**Note:** In environments with CLI access you can also restart the service using the following command:

```
sudo systemctl restart sc4s
```

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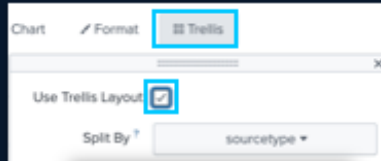


## Example search to check events:

```
index=* host=192* sourcetype=*cisco*  
| timechart count by sourcetype
```

# null\_queue

```
index=main host=192* sourcetype=*cisco*  
| timechart count by sourcetype
```



Enable the trellis layout and split by **sourcetype**



We are currently getting data from:

- ~~fws~~
- asa

**fws** has dropped away thanks to our filter

# Check Labs Dashboard

